Nicholas Harrison

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Skills

CAD: SolidWorks, Siemens NX, Fusion 360
Analysis: Ansys Mechanical, Thermal Desktop (Icepak), SolidWorks FEA, VI-CarRealTime, Foxglove
Programming: Python, C++, MATLAB, Simulink

Experience

Driverless and Vehicle Dynamics Member | University of Toronto Formula Racing Sept 2026 - Present

- Implemented a custom algorithm to detect and filter duplicate cones from LiDAR and camera data
- Developed and implemented key metrics used to evaluate the performance of the car during simulation

Teaching Assistant | University of Toronto | Department of Mechanical Engineering Sept 2026 - Present

- Providing feedback on student assignments and term project submissions done in SolidWorks.

Vehicle Dynamics and Control Research Intern | Intelligent Mobility and Robotics Lab Jan 2026 – Aug 2026

- Conducted research in collaboration with UWaterloo's Dept. of Mechanical and Mechatronics Engineering and UNB master's students, focused on developing a physics-informed predictive speed and braking advisory system for heavy trucks.
- Developed the foundation of a vehicle dynamics control algorithm to improve the safe operation of Class-8 trucks.
- Paper accepted into the 2026 International Conference on Vehicle Engineering and Intelligent Control (VEIC). SAE
- Developing a 7-DOF model for autonomous adaptive suspension.

Project Coordinator | EllisDon May 2026 - July 2026

- Assisted the Project Manager in planning and executing the project.
- Monitored project performance using KPIs and progress tracking tools, identified schedule variances and potential risks.
- Administered change control processes, documented scope changes and assessed schedule and cost impacts.

Drivetrain Lead – UNB Formula Racing Sept 2025 – July 2026

- Developed CAD models and assemblies for drivetrain manufacturing in SolidWorks.
- Performed finite element analysis (FEA) on mounting structures and shafts to verify structural integrity and identify areas where material is unnecessary.
- Applied bearing load equations to determine the maximum dynamic radial load capacity of the mount bearings, accounting for a factor of safety and desired bearing life.

Education

MEng | University of Toronto 2026-2028
Robotics, Dept. of Mechanical and Industrial Engineering

BSc Mechanical Engineering | University of New Brunswick | First Division 2026
Mechatronics option, Minor in Math

Projects

Quadcopter Drone: Custom ESP32 flight controller-based drone, flown using an Xbox controller.
Fixed-Wing Drone: Custom build using a foamboard chassis, 3D-printed wing frame, and flight controller.
Flappy Bird Bot: Designed a model predictive controller for Flappy Bird, which allows the game to play itself.
Mobile RC Robot: Built a robot that can be driven around and perform certain tasks.

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